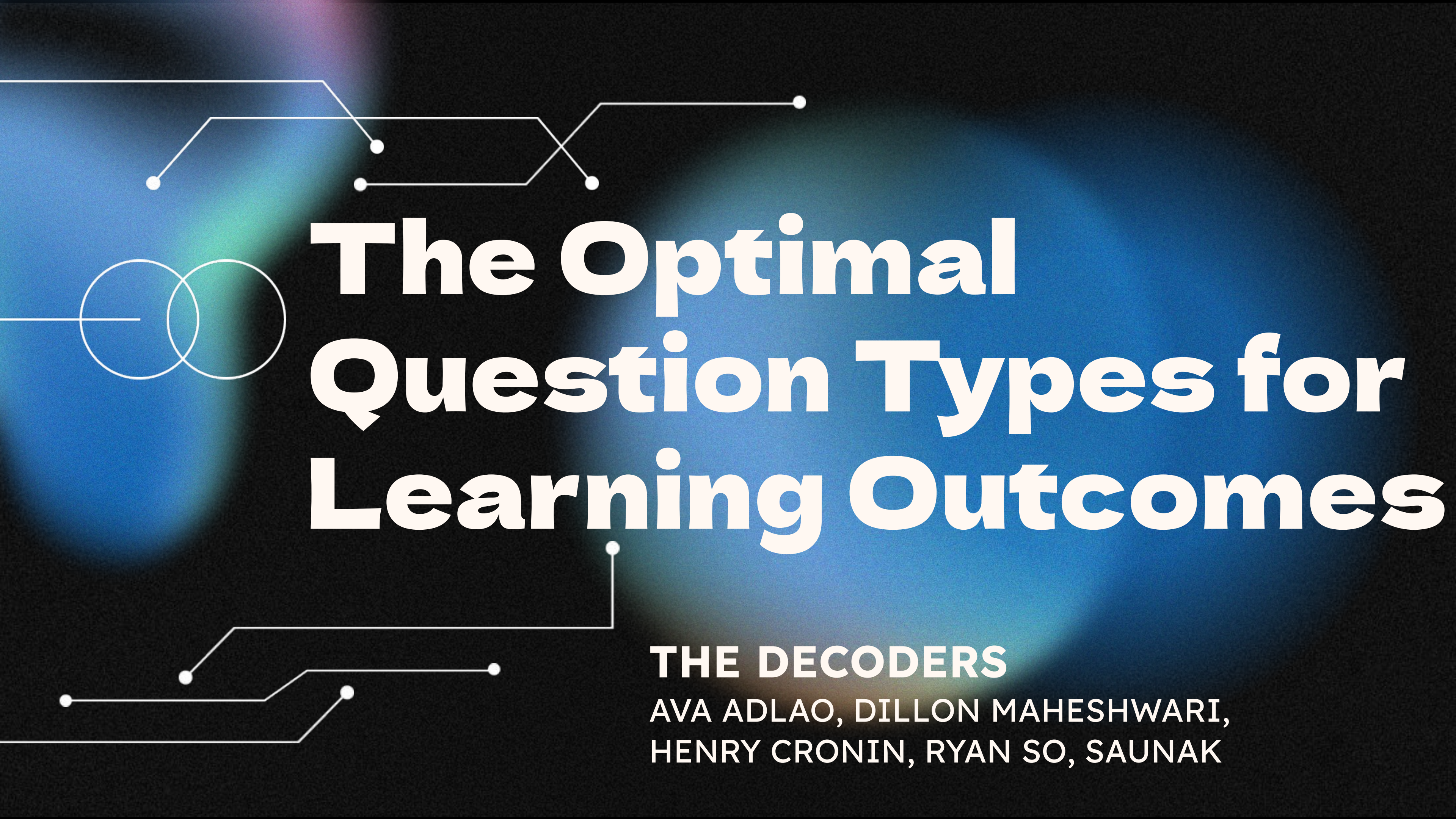




The Optimal Question Types for Learning Outcomes

THE DECODERS

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QUESTION TYPES

mcq

Model comparisons typically involve comparing a complex model to a simple model. Which two models are being compared in the M

A

complex model: $\text{later}_i = b_0 + b_1\text{Dogs}_i + b_2\text{based}_i + b_3\text{Dogs}_i * \text{based}_i + e_i$
simple model: $\text{later}_i = b_0 + e_i$

B

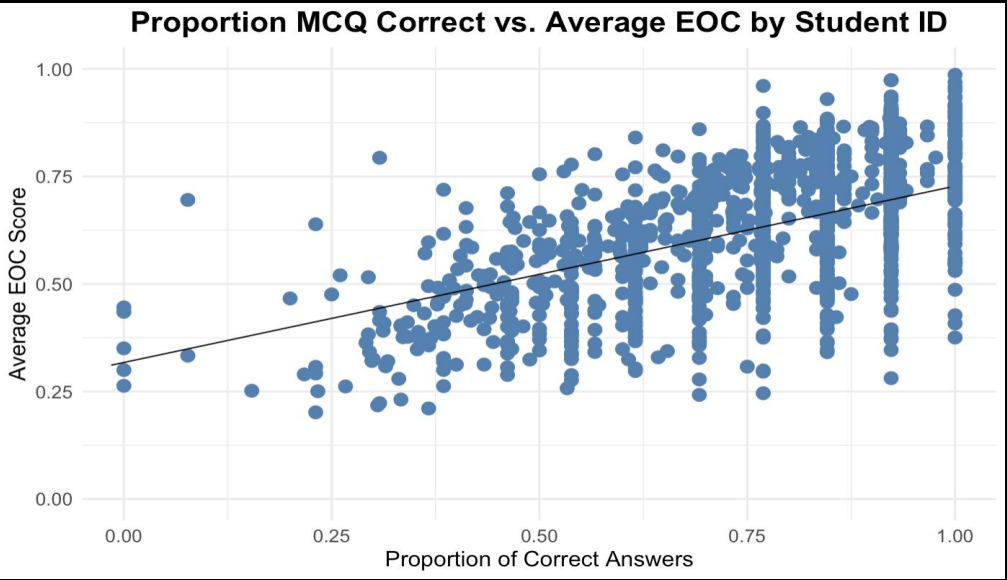
complex model: $\text{later}_i = b_0 + b_1\text{Dogs}_i + b_2\text{based}_i + e_i$
simple model: $\text{later}_i = b_0 + e_i$

C

complex model: $\text{later}_i = b_0\text{Dogs}_i * \text{based}_i + e_i$
simple model: $\text{later}_i = b_0 + e_i$

D

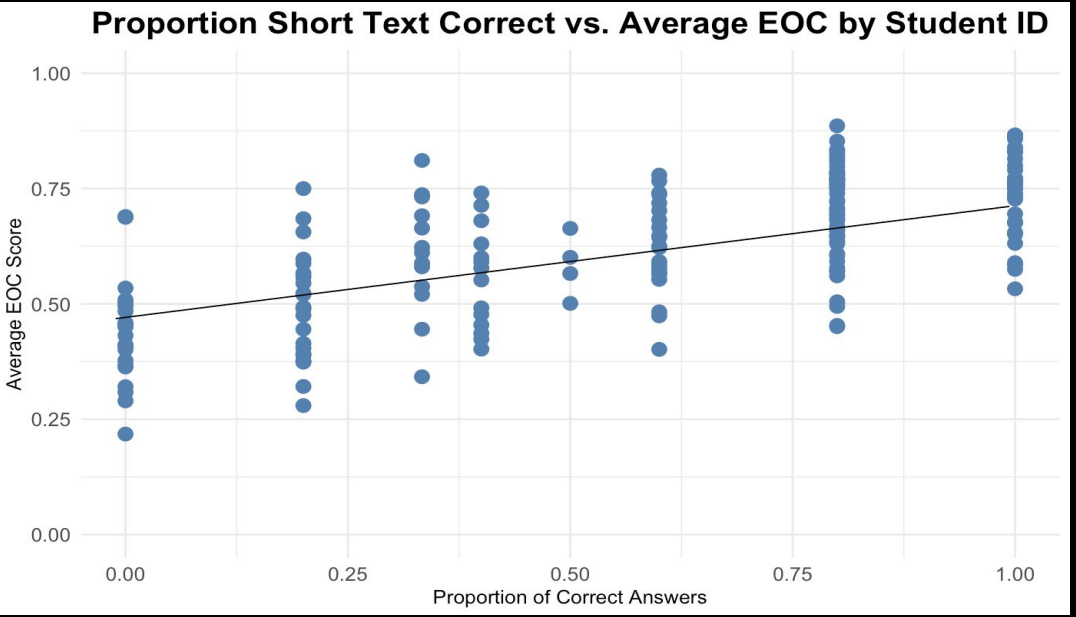
complex model: $\text{later}_i = b_0 + b_1\text{Dogs}_i + b_2\text{based}_i + b_3\text{Dogs}_i * \text{based}_i + e_i$
simple model: $\text{later}_i = b_0 + b_1\text{Dogs}_i + b_2\text{based}_i + e_i$



shorttext

	Condition	Age	Wt	Wt2
1	Uninformed	35	136	135.8
2	Uninformed	45	162	161.8
3	Informed	52	117	116.8
4	Informed	29	184	182.8
5	Uninformed	38	134	136.6
6	Informed	39	189	183.2

Notice the numbers 1 through 6 down the left column. What do these n



association

Which features of the boxplot go with which statistical summary?

The lowest point of the bottom whisker

The top line of the box (this is called the top hinge)

The bottom line of the box (this is called the bottom hinge)

The height of the box

The highest point of the top whisker

The middle line of the box

Max (Q4) = 196

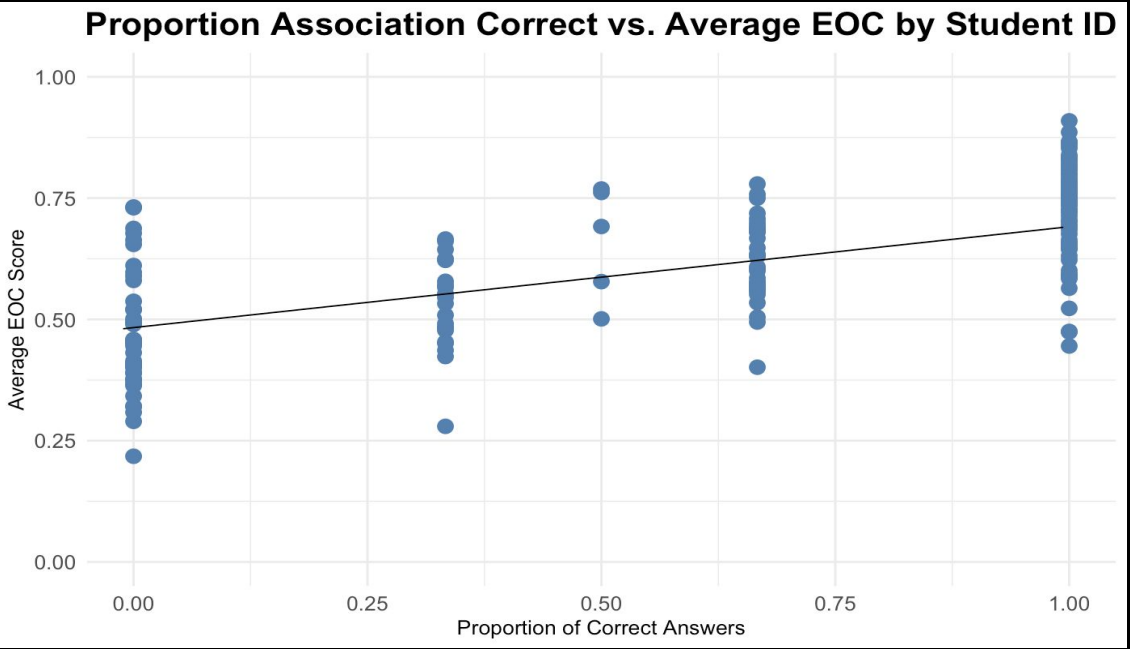
Q3 = 161.5

IQR = Q3 - Q1 = 31.5

Median (Q2) = 145

Min (Q0) = 90

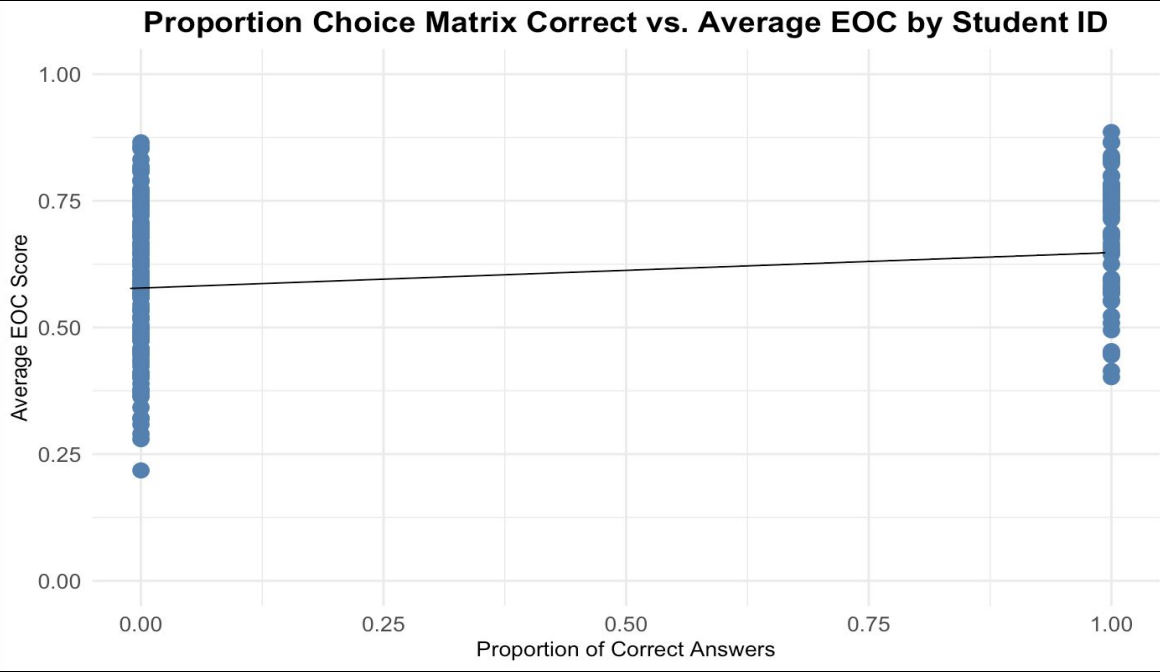
Q1 = 130



choice matrix

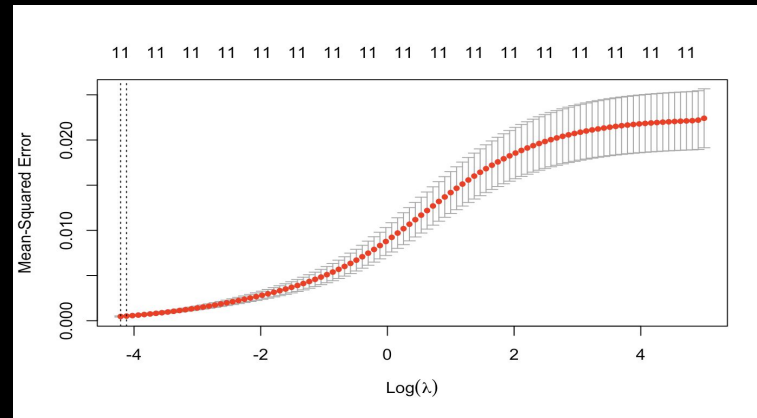
Which of these are values and which are variables?

		Value	Variables	Neither
A	Condition	☐	☐	☐
B	Wt2	☐	☐	☐
C	45	☐	☐	☐
D	Housekeeper	☐	☐	☐
E	Uninformed	☐	☐	☐
F	135.8	☐	☐	☐



RIDGE REGRESSION

cross-validation

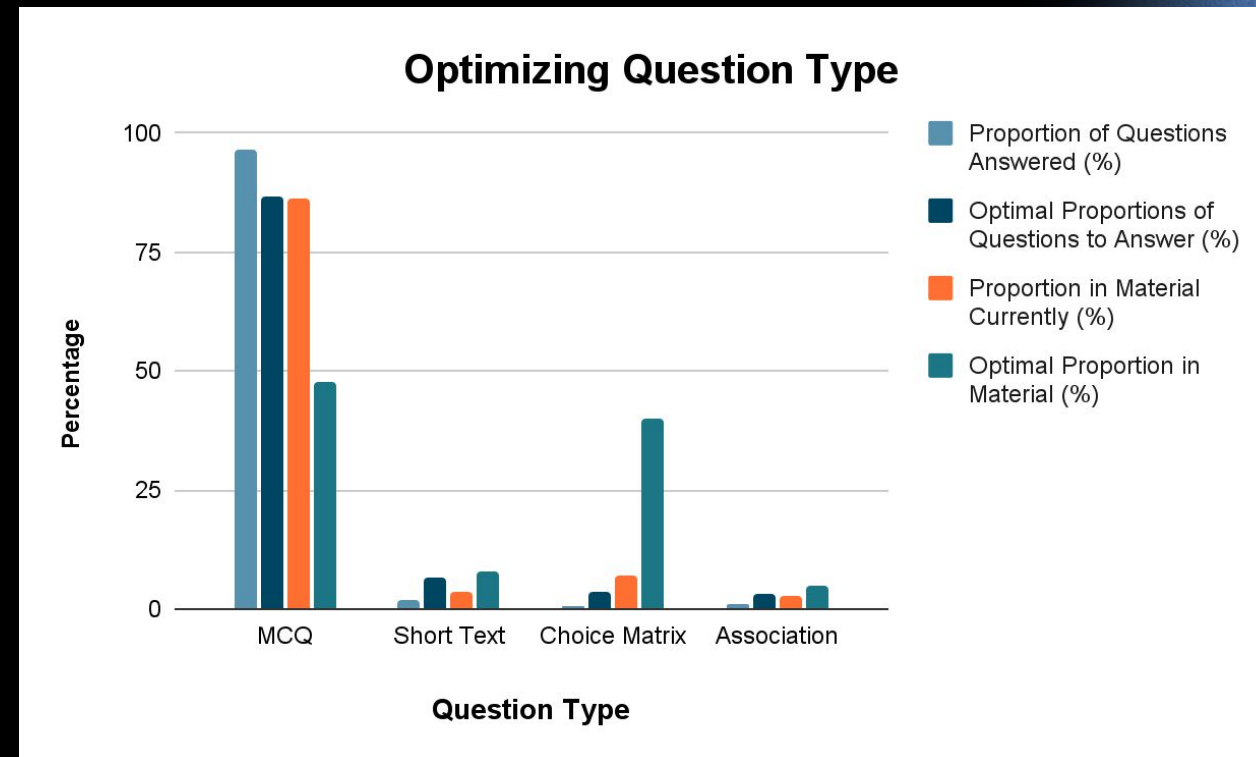


- Best tuning parameter: lambda corresponding to minimum error OR to no more than 1 SE above the minimum

coefficients

Predictors	Coefficient
(Intercept)	0.0941372310
mcq	0.1576076934
association	0.0121430822
choicematrix	0.0069734997
shorttext	0.0055129621
mcqA	0.0258571838
mcQC	0.0264058108
mcQS	0.0205737940
AssC	-0.0002865324
AssS	-0.0060786797
ChoiceS	-0.0134995579

Optimal Proportion of Each Question Type



question type	proportion of questions answered (%)	optimal proportions of questions to answer (%)	proportion in material currently (%)	optimal proportion in material (%)	proportion in test (%)	correlation
MCQ	96.47	86.48	86.11	47.56	94.26	0.62149
Short Text	01.99	6.66	3.8	7.84	0	0.7219
Choice Matrix	0.43	3.83	7.21	39.85	0.68	0.6971
Association	1.12	3.03	2.83	4.74	5.06	0.2898

- lambda corresponding to minimum error